Exploratory Data Analysis

Kaggle – Weather Type Classification dataset

Abhay Burye · Jalesh Kumar · Uttam K

This document explores the raw dataset **before** any cleaning or modelling: its structure, data quality, the distribution of each feature, how each feature relates to the weather type, and how features relate to each other. All figures and tables are produced by **eda.py** in the project repository.

Key findings at a glance

- **13,200 rows, 10 features, 4 perfectly balanced classes** (3,300 each). No missing values and no duplicate rows.
- **Data-quality problems:** about 3% of rows have humidity or precipitation above 100% (impossible), 2.7% have temperatures above 50 °C, and about 7% have atmospheric pressure outside 950–1050 hPa. The data is at least partly synthetic.
- **Every feature differs strongly between weather types** (all p-values effectively 0). Temperature ($\eta^2 = 0.52$) and precipitation ($\eta^2 = 0.46$) separate the classes best; pressure is weakest ($\eta^2 = 0.07$).
- **Some rules are almost deterministic:** clear sky → always Sunny (2,139 of 2,139 rows); 94% of Snowy rows are in Winter; Snowy is rare at the coast (120 of 3,571 coastal rows).
- **Rainy vs Snowy** look alike on humidity, precipitation and visibility; **temperature** is what tells them apart. **Cloudy vs Rainy** have the same temperatures; **precipitation** tells them apart.
- **Features are moderately correlated** (strongest: humidity–precipitation r = 0.64), so there is some redundancy but no near-duplicate feature.

1. Structure of the data

Variable	Type	Range in raw data	Notes
Temperature	Numeric (°C)	−25 to 109	Values above 50 °C are implausible
Humidity	Numeric (%)	20 to 109	Values above 100% are impossible
Wind Speed	Numeric (km/h)	0 to 48.5	Right-skewed
Precipitation (%)	Numeric (%)	0 to 109	Values above 100% are impossible
Atmospheric Pressure	Numeric (hPa)	800 to 1,199	Real sea-level pressure is roughly 870–1,085
UV Index	Numeric (integer)	0 to 14	
Visibility (km)	Numeric (km)	0 to 20	Right-skewed
Cloud Cover	Categorical	4 levels	clear, partly cloudy, cloudy, overcast
Season	Categorical	4 levels	Winter is over-represented (5,610 rows, 43%)
Location	Categorical	3 levels	inland 4,816 · mountain 4,813 · coastal 3,571
Weather Type	Target	4 classes	Sunny, Cloudy, Rainy, Snowy – 3,300 each

Table 1. Variables in the dataset

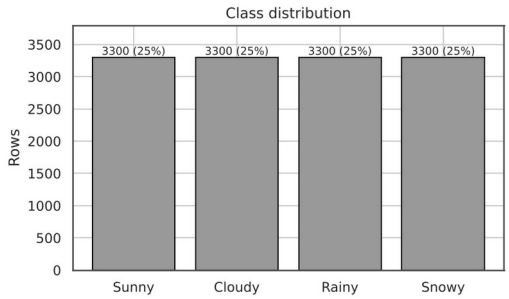


Figure 1. The target classes are perfectly balanced, so accuracy is a fair headline metric

2. Data quality

There are **no missing values and no duplicate rows**. However, several values are physically impossible or implausible:

Check	Rows	% of data	What we did
Humidity > 100%	416	3.2%	Set to missing, filled with training median
Precipitation > 100%	392	3.0%	Set to missing, filled with training median
Temperature > 50 °C	356	2.7%	Kept (no clear cut-off); noted as a limitation
Pressure < 950 hPa	442	3.3%	Kept; tree models are robust to extreme values
Pressure > 1,050 hPa	458	3.5%	Kept; tree models are robust to extreme values

Table 2. Impossible or implausible values in the raw data

These patterns (round ceilings at 109, uniform spreads of extreme values) suggest that noise was injected into an otherwise generated dataset. This matters for interpretation: model scores here may not transfer to real weather records.

3. Distribution of each numeric feature

Feature	Mean	Median	Std dev	Q1	Q3	Skew	Kurtosis	IQR outliers
Temperature	19.13	21.00	17.39	4.0	31.0	0.22	0.59	92
Humidity	68.71	70.00	20.19	57.0	84.0	−0.40	−0.34	0
Wind Speed	9.83	9.00	6.91	5.0	13.5	1.36	3.26	404
Precipitation (%)	53.64	58.00	31.95	19.0	82.0	−0.15	−1.35	0
Atm. Pressure	1005.83	1007.65	37.20	994.8	1016.8	−0.29	12.78	927
UV Index	4.01	3.00	3.86	1.0	7.0	0.90	−0.36	0
Visibility (km)	5.46	5.00	3.37	3.0	7.5	1.23	2.52	383

Table 3. Descriptive statistics of the raw numeric features (outliers by the 1.5 × IQR rule)

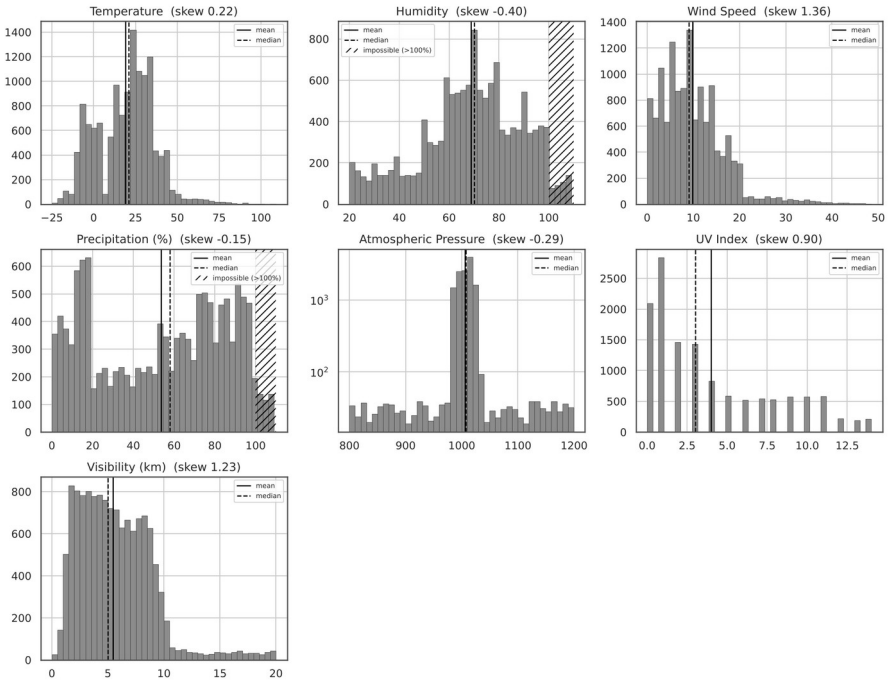


Figure 2. Histograms with mean (solid) and median (dashed); hatched areas are impossible values. Pressure uses a log scale.

What the shapes tell us:

- **Wind speed, UV index and visibility are right-skewed** (mean above median), with long right tails. The median is the better summary of a typical value, which is why we impute with the median.
- **Atmospheric pressure** is tightly packed around 1,005–1,020 hPa but has very heavy tails (kurtosis 12.8): 927 IQR outliers spread almost uniformly from 800 to 1,200 hPa, a sign of injected noise.
- **Precipitation** is bimodal: a cluster below 20% (dry weather) and a broad band from 50% to 100% (wet weather). This already hints that it will separate dry from wet classes.
- **Temperature** has a bump below 0 °C (snowy days) and a long tail above 50 °C (implausible values).
- **UV index** takes integer values and is concentrated at 0–2, with a flat spread from 5 to 11.

4. How features relate to the weather type

Feature	Sunny	Cloudy	Rainy	Snowy
Temperature (°C)	32.4	22.8	22.8	−1.5
Humidity (%)	51.4	66.5	78.4	78.5
Wind Speed (km/h)	6.1	8.6	13.7	11.0
Precipitation (%)	25.0	40.3	74.8	74.6
Atm. Pressure (hPa)	1017.9	1010.2	1004.2	991.1
UV Index	7.8	3.6	2.7	2.0
Visibility (km)	7.6	7.1	3.6	3.6

Table 4. Mean of each feature by weather type

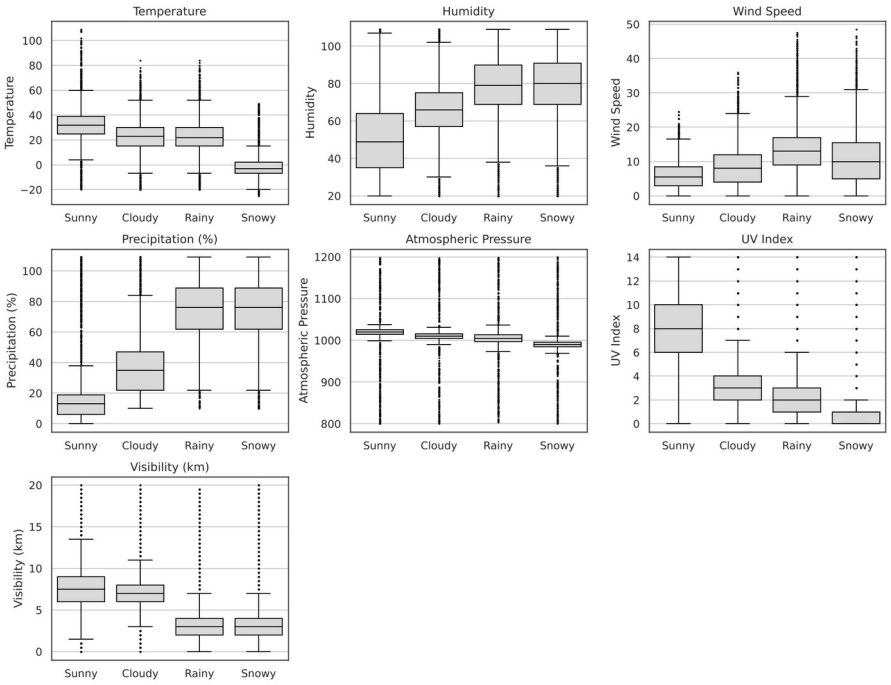


Figure 3. Box plots of each numeric feature by weather type

Sunny days are hot, dry, bright (high UV, long visibility) and high-pressure. **Rainy** and **Snowy** days are both humid, wet and low-visibility; the big difference is **temperature** (about 23 °C vs −1.5 °C). **Cloudy** days sit between Sunny and Rainy on almost every feature.

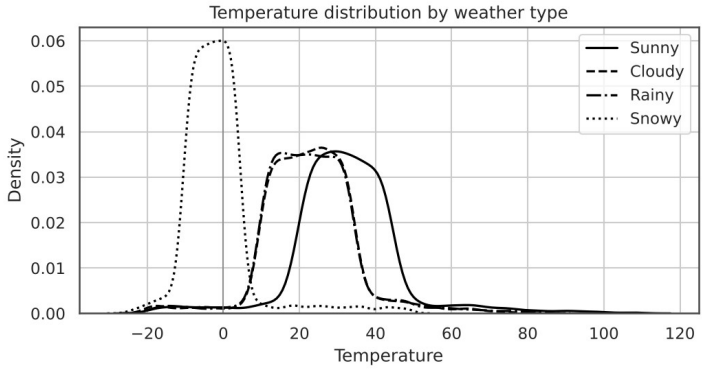


Figure 4. Temperature density by class: Snowy is clearly separated, Cloudy and Rainy overlap almost completely

Figure 4 shows why no single feature is enough. Temperature isolates Snowy, but Cloudy and Rainy have nearly identical temperature curves; precipitation and visibility separate those two instead. A model has to **combine** features, which favours tree-based models over a linear one.

5. Categorical features and weather type

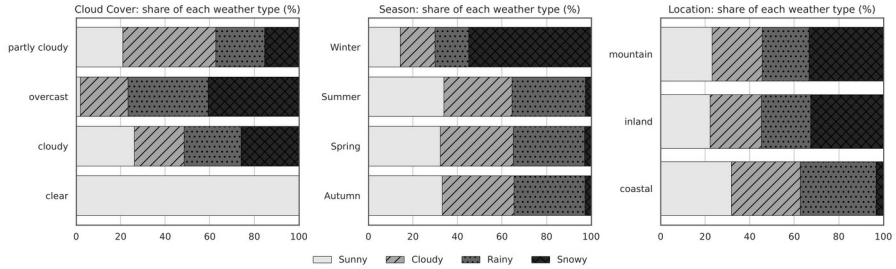


Figure 5. Share of each weather type within each category (rows sum to 100%)

- **Cloud cover:** a **clear** sky is **always Sunny** (2,139 of 2,139 rows). Overcast skies are mostly Snowy (41%) or Rainy (36%), and almost never Sunny (2%).
- **Season: Winter** holds **94% of all Snowy rows** (3,086 of 3,300). In the other seasons Snowy is only about 3%, and the other three types are evenly split.
- **Location:** Snowy is rare at the **coast** (120 of 3,571 rows, 3%) but makes up a third of inland and mountain rows.

6. Statistical tests of association

To check formally that each feature is related to the weather type, we used **one-way ANOVA** (difference in means), the **Kruskal–Wallis test** (a rank-based check that does not assume normality, useful given the skew) and the **chi-square test of independence** for categorical features. Effect sizes show *how strong* each relationship is: **η^2** is the share of a feature's variance explained by the weather type, and **Cramér's V** runs from 0 (no association) to 1 (perfect association).

Feature	Test	Statistic	p-value	Effect size
Temperature	ANOVA F / Kruskal–Wallis H	4,798 / 6,837	$< 10^{-300}$	$\eta^2 = 0.52$
Precipitation (%)	ANOVA F / Kruskal–Wallis H	3,777 / 5,542	$< 10^{-300}$	$\eta^2 = 0.46$
UV Index	ANOVA F / Kruskal–Wallis H	2,327 / 5,311	$< 10^{-300}$	$\eta^2 = 0.35$
Humidity	ANOVA F / Kruskal–Wallis H	1,911 / 4,035	$< 10^{-300}$	$\eta^2 = 0.30$
Visibility (km)	ANOVA F / Kruskal–Wallis H	1,928 / 6,371	$< 10^{-300}$	$\eta^2 = 0.30$
Wind Speed	ANOVA F / Kruskal–Wallis H	877 / 2,514	$< 10^{-300}$	$\eta^2 = 0.17$
Atm. Pressure	ANOVA F / Kruskal–Wallis H	330 / 6,505	7×10^{-207}	$\eta^2 = 0.07$
Cloud Cover	Chi-square (df = 9)	9,407	$< 10^{-300}$	V = 0.49
Season	Chi-square (df = 9)	4,695	$< 10^{-300}$	V = 0.34
Location	Chi-square (df = 6)	1,235	1×10^{-263}	V = 0.22

Table 5. Association between each feature and the weather type (sorted by effect size)

All p-values are essentially zero, as expected with 13,200 rows: even small effects become "significant" in a large sample, so the **effect sizes are more informative**. Temperature and precipitation are the strongest single predictors, and cloud cover is the strongest categorical one. Pressure has a tiny η^2 (0.07) because its extreme noisy values swamp the mean, yet its Kruskal–Wallis H is large: on ranks, which ignore the extremes, pressure does separate the classes.

7. Relationships between features

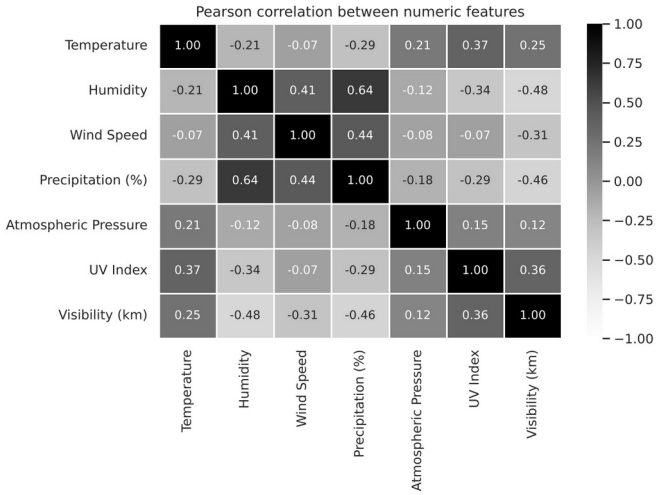


Figure 6. Pearson correlation matrix of the numeric features

- **Humidity and precipitation** are the most correlated pair (r = 0.64): wet weather is humid weather.
- **Visibility** falls as humidity (r = −0.48) and precipitation (r = −0.46) rise.
- **Wind speed** rises with humidity (r = 0.41) and precipitation.
- **UV index** rises with temperature (r = 0.37) and visibility (r = 0.36).
- No pair is above |r| = 0.7, so there is no serious multicollinearity, and every feature carries some information of its own.

8. What the EDA means for modelling

1. **Accuracy is a fair metric** because the classes are perfectly balanced, and a **stratified split** keeps that balance in train and test.
2. **Clean the impossible percentages** (set to missing, impute with the training **median**, which suits the skewed features) and do it inside the pipeline to avoid leakage.
3. **Tree-based models suit this data:** the classes are separated by thresholds and combinations (clear sky → Sunny; cold + wet → Snowy; warm + wet → Rainy), which is exactly how a decision tree splits. This is why our Decision Tree (90.8%) and Random Forest (91.2%) score so close to each other; a linear Logistic Regression reaches only 86.7%.
4. **Expect a ceiling around 91%:** the injected noise (impossible temperatures and pressures, overlap between Cloudy and Rainy) means some rows cannot be classified correctly by any model.
5. **No scaling is needed** for the Decision Tree and Random Forest, because tree splits are unaffected by the scale of a feature.